WEEK 11 – AUTOMOBILE DATA CLEANING, TRANSFORMATION AND FEATURE ENGINEERING USING POWER BI

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1. INTRODUCTION

The Week 11 project focuses on Automobile Data Cleaning, Transformation, and Feature Engineering using Microsoft Power BI. Data cleaning is an important step in data analysis because raw datasets may contain missing values, duplicate records, incorrect data types, and invalid entries.

In this project, an automobile dataset is used to understand different vehicle specifications, engine performance, fuel efficiency, and pricing information. Power Query Editor is used to clean and transform the raw data into a structured format. New calculated columns are also created to improve the analytical value of the dataset.

The main aim of this project is to prepare a reliable and well-organized automobile dataset for further analysis and visualization.

2. OBJECTIVES OF THE PROJECT

The major objectives of this project are:

- To import automobile data into Microsoft Power BI.
- To understand the structure and properties of the dataset.
- To identify and handle missing and invalid values.
- To detect and remove duplicate records.
- To correct errors in numerical columns.
- To convert columns into appropriate data types.

- To create new calculated columns using Feature Engineering.
- To develop DAX measures for calculating important statistics.
- To prepare a clean dataset for further visualization and reporting.

3. DATASET DESCRIPTION

The Automobile dataset contains information about different types of vehicles and their technical specifications. It includes details about automobile manufacturers, fuel types, engine characteristics, vehicle dimensions, horsepower, mileage, and prices.

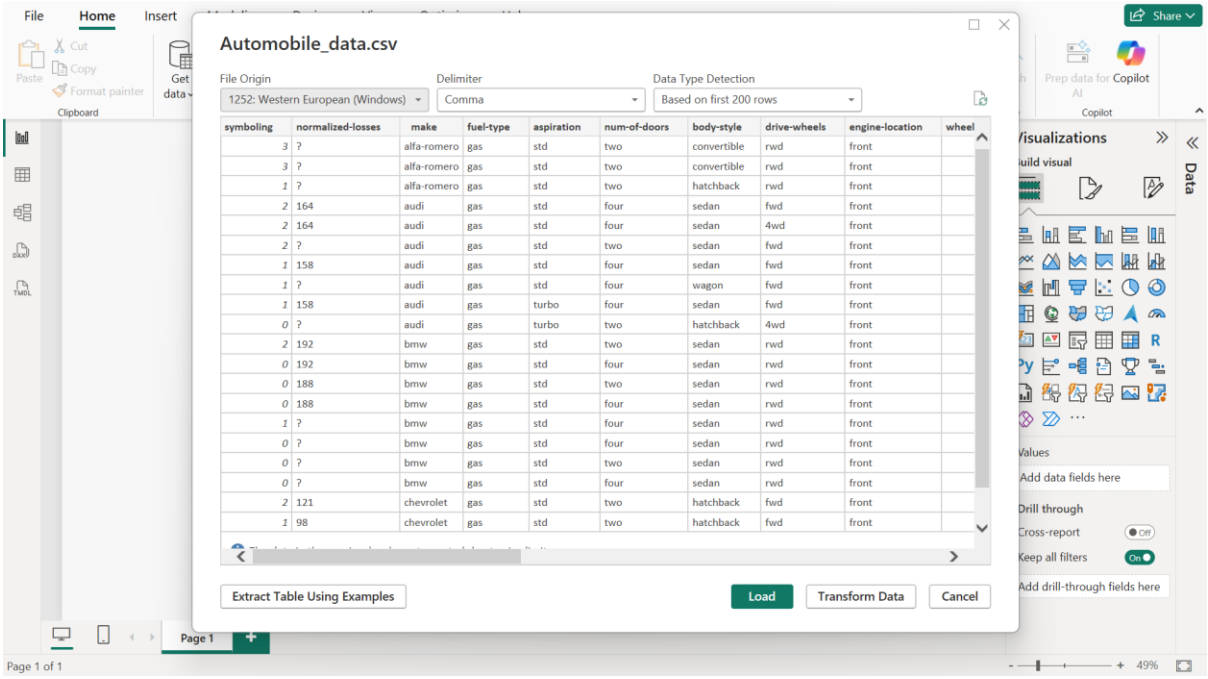
The dataset was provided in CSV format and imported into Power BI for data preparation.

4. DATA IMPORTATION

The automobile dataset was imported into Microsoft Power BI using the Get Data option. The Text/CSV option was selected to load the dataset.

After selecting the CSV file, the Transform Data option was used to open Power Query Editor. This editor provides various features for cleaning, modifying, and preparing the dataset before loading it into the Power BI environment.

The dataset was examined to identify the available columns, their data types, and the presence of missing or invalid values.



5. DATA CLEANING USING POWER QUERY

Data cleaning is the process of identifying and correcting errors and inconsistencies in a dataset. In this project, Power Query Editor was used to perform several data cleaning operations.

5.1 Handling Missing Values

Some columns in the automobile dataset contained missing values represented by the symbol '?'. These values were identified as invalid entries because they could affect numerical calculations and analysis.

The Replace Values and Custom Column options were used to handle these missing entries. Invalid

values were converted into null values so that they could be processed correctly.

The following columns required special attention:

- Normalized Losses
- Bore
- Stroke
- Horsepower
- Peak RPM
- Price

5.2 Error Handling

Errors were identified in certain numerical columns when converting the data types. To solve this problem, new custom columns were created using Power Query formulas.

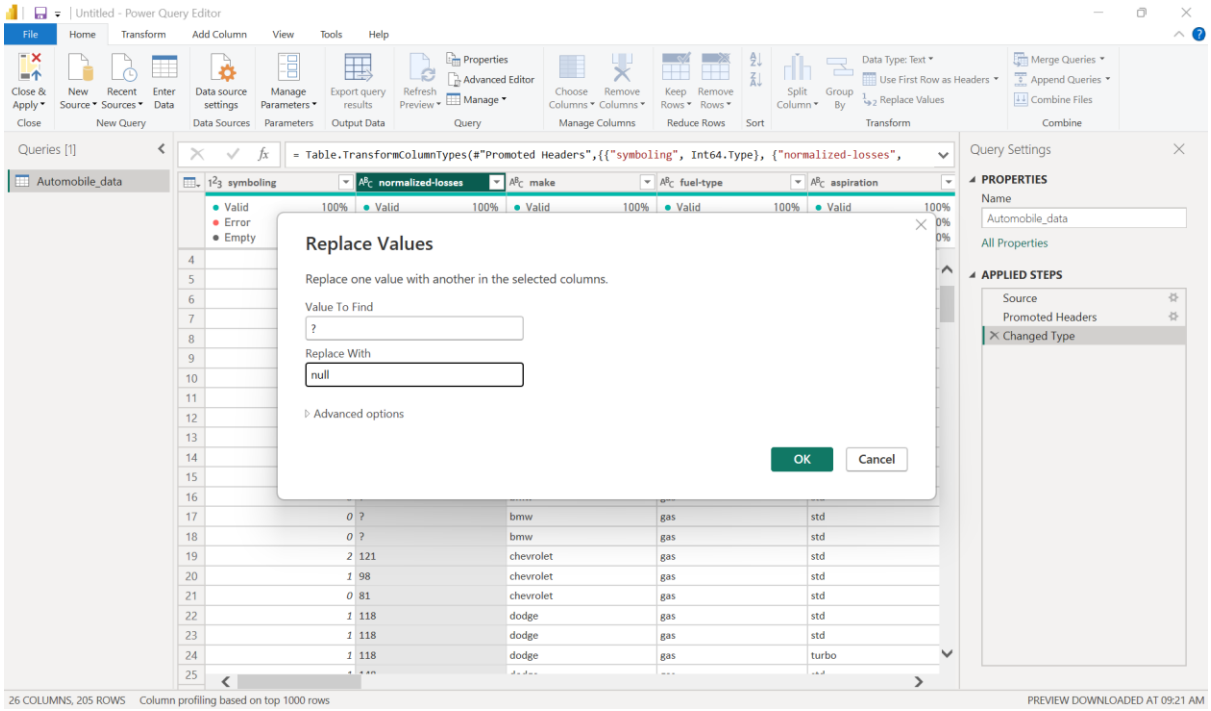
The TRY and OTHERWISE functions were used to convert valid values into numbers and replace invalid entries with null values.

This method helped create clean numerical columns without unnecessary conversion errors.

5.3 Removing Unnecessary Columns

After creating and validating the cleaned columns, the original columns containing invalid values were removed where appropriate. The cleaned columns were renamed using their original column names.

This helped maintain a clear and organized dataset structure.



6. DATA TYPE CONVERSION

Data type conversion is an important part of data preparation. Each column was assigned a suitable data type according to the nature of its values.

The following data types were used:

- Whole Number: Price, Horsepower, Engine Size, Peak RPM, City MPG, Highway MPG and Curb Weight.
- Decimal Number: Bore and Stroke.
- Text: Manufacturer, Fuel Type, Body Style, Drive Wheels and Fuel System.

Correct data types help improve calculation accuracy and make the dataset suitable for further analysis.

7. DUPLICATE DATA REMOVAL

Duplicate records can affect the accuracy of data analysis and produce incorrect results. Therefore, the Remove Duplicates feature available in Power Query was used to check and remove duplicate records.

The dataset was checked using the available columns to identify completely duplicated rows. This process helped maintain data consistency and avoid repeated records.

8. FEATURE ENGINEERING

Feature Engineering is the process of creating new useful columns from existing data. It helps improve the quality of analysis and provides additional information for understanding the dataset.

In this project, three new calculated columns were created using the Custom Column option in Power Query.

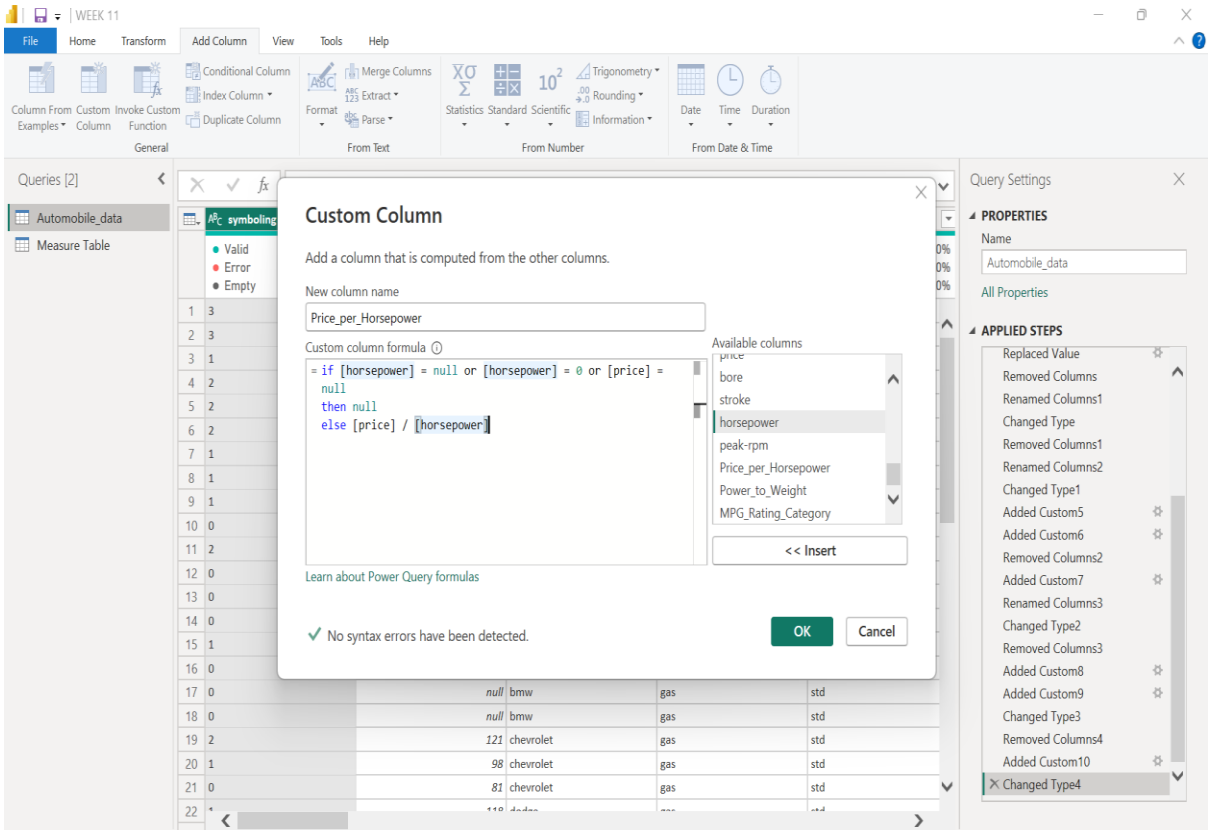
8.1 Price per Horsepower

A new column named Price_per_Horsepower was created to calculate the price of a vehicle relative to its engine horsepower.

Formula:

Price per Horsepower = Price / Horsepower

This feature helps understand the relationship between vehicle pricing and engine performance.



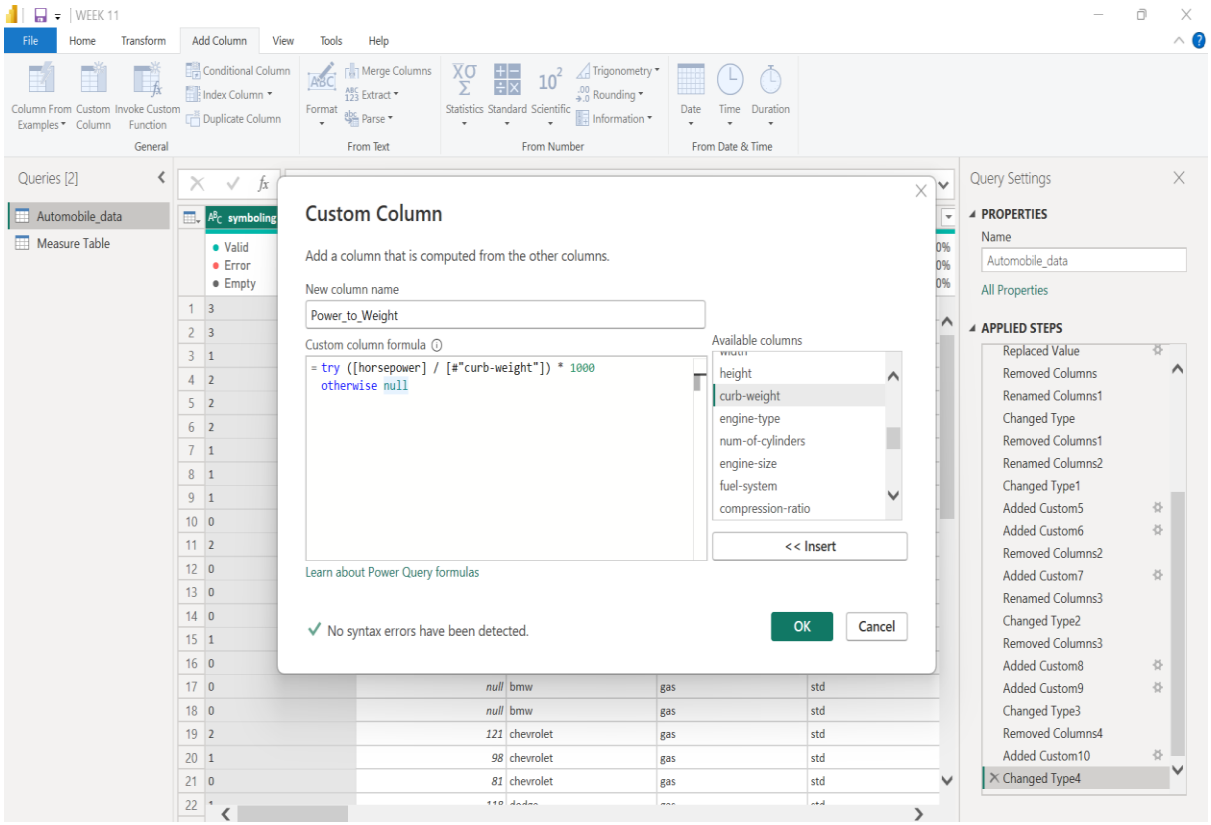
8.2 Power-to-Weight Ratio

A new column named Power_to_Weight was created using the horsepower and curb-weight columns.

Formula:

Power-to-Weight Ratio = (Horsepower / Curb Weight) × 1000

This calculation provides an additional measure of the relationship between engine power and vehicle weight.



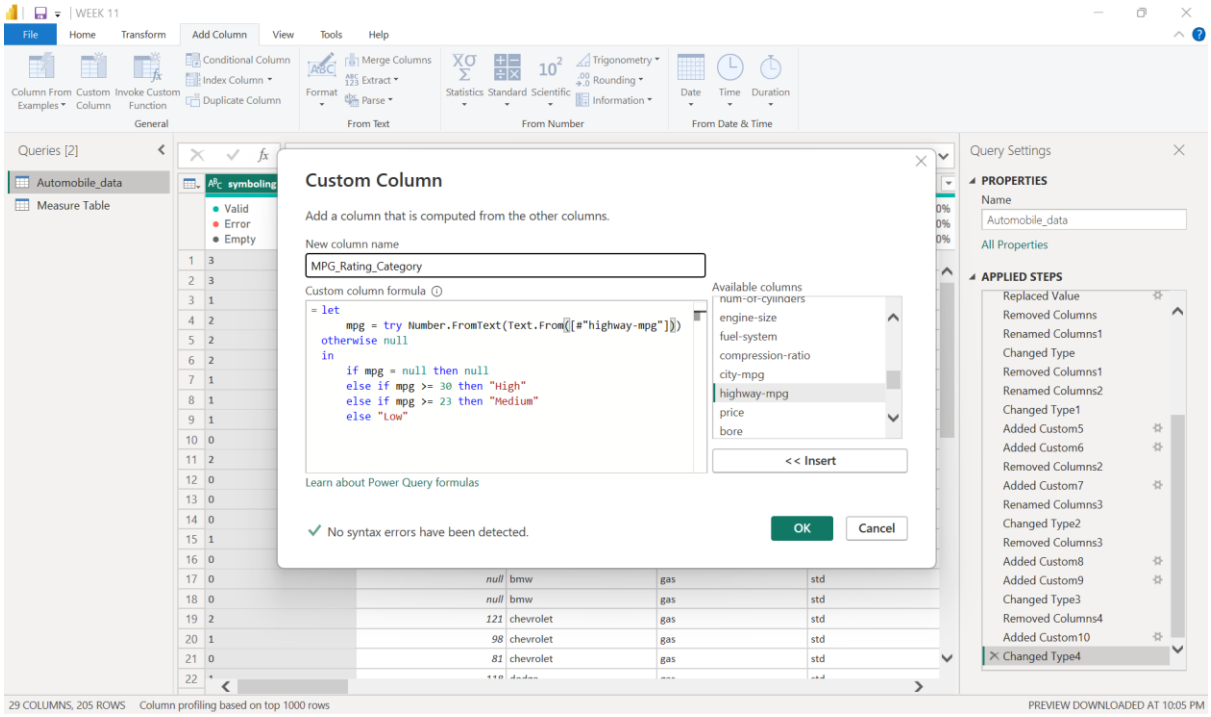
8.3 MPG Rating Category

A new column named MPG_Rating_Category was created to classify vehicles according to their highway fuel efficiency.

The vehicles were divided into three categories:

- High: Highway MPG greater than or equal to 30.
- Medium: Highway MPG between 23 and 29.
- Low: Highway MPG below 23.

This classification makes it easier to compare vehicles based on their fuel efficiency.



9. DATA TRANSFORMATION AND VALIDATION

After completing the data cleaning and feature engineering operations, the transformed dataset was checked for consistency.

The newly created columns were verified to ensure that the calculated values were generated correctly. Data types were also reviewed to avoid calculation errors.

The dataset was examined for remaining errors, invalid entries, and missing values. The final transformed data was prepared for loading into Power BI.

10. DAX MEASURES

DAX (Data Analysis Expressions) is a formula language used in Power BI to perform calculations and create meaningful summaries from data.

The following DAX measures were planned for automobile data analysis:

10.1 Average Price

Calculates the average price of automobiles in the dataset.

Average Price = AVERAGE(Automobile_data[price])

10.2 Average Horsepower

Calculates the average horsepower of the vehicles.

Average Horsepower = AVERAGE(Automobile_data[horsepower])

10.3 Average Engine Size

Calculates the average engine size of the automobiles.

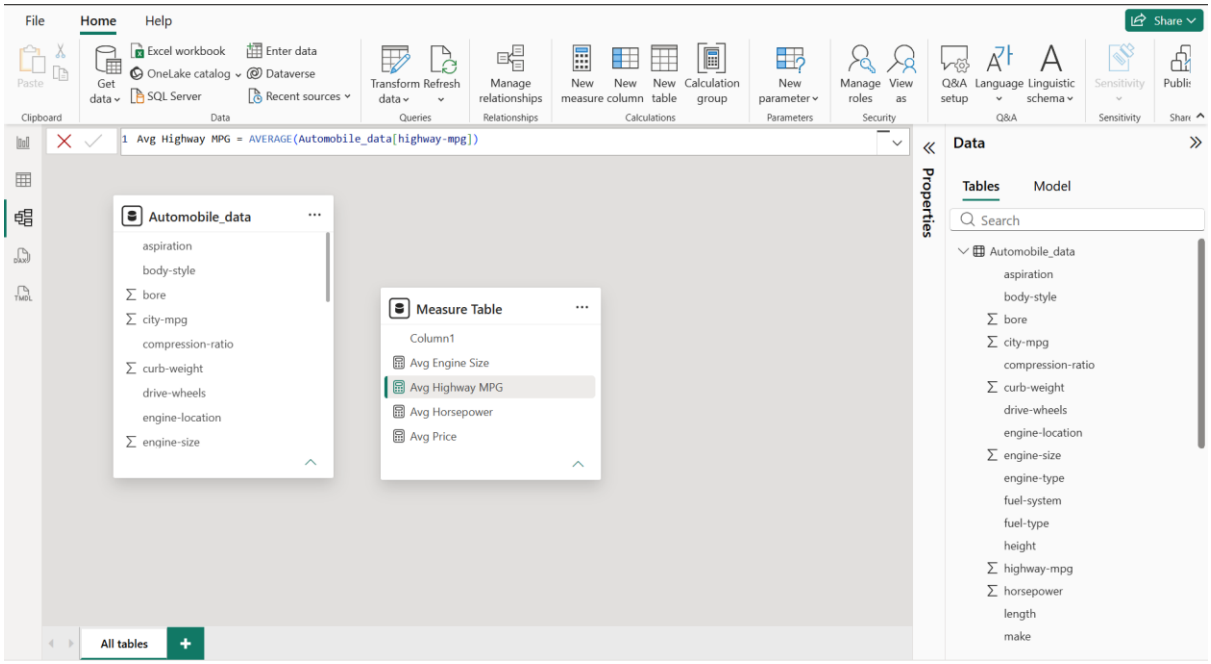
Average Engine Size = AVERAGE(Automobile_data[engine-size])

10.4 Average Highway MPG

Calculates the average highway fuel efficiency of the vehicles.

Average Highway MPG = AVERAGE(Automobile_data[highway-mpg])

These measures help summarize important automobile specifications and support further data analysis.



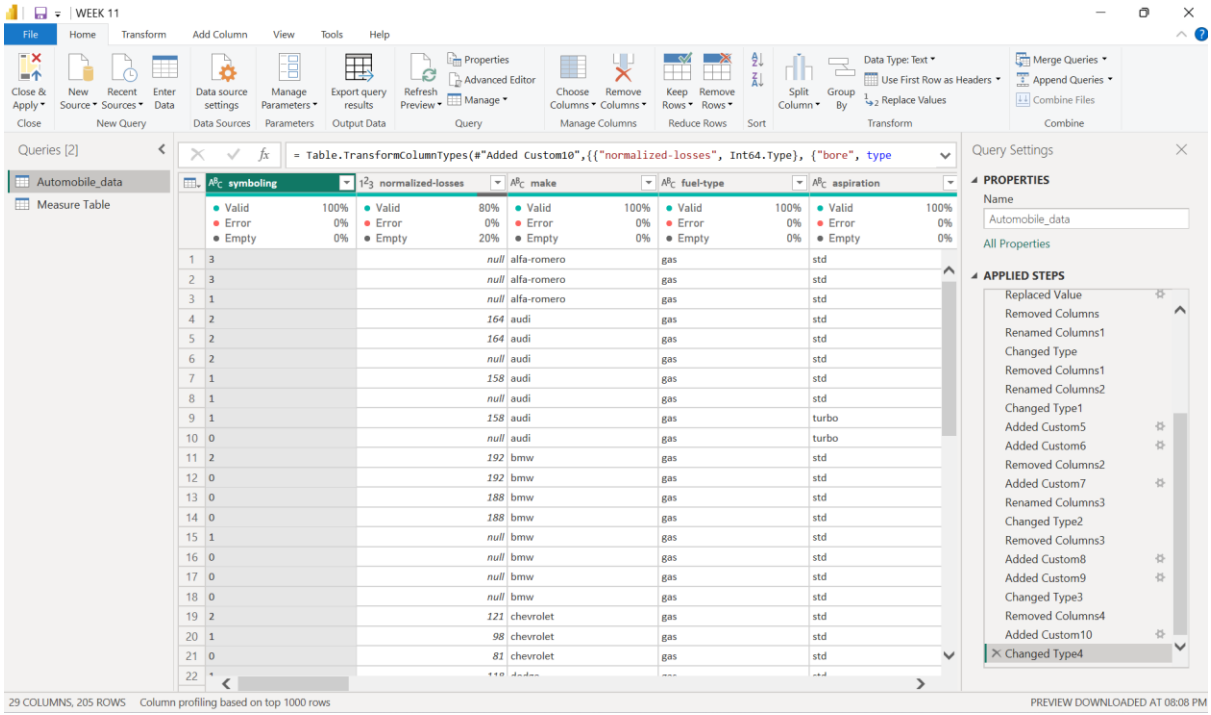
11. FINAL DATA PREPARATION

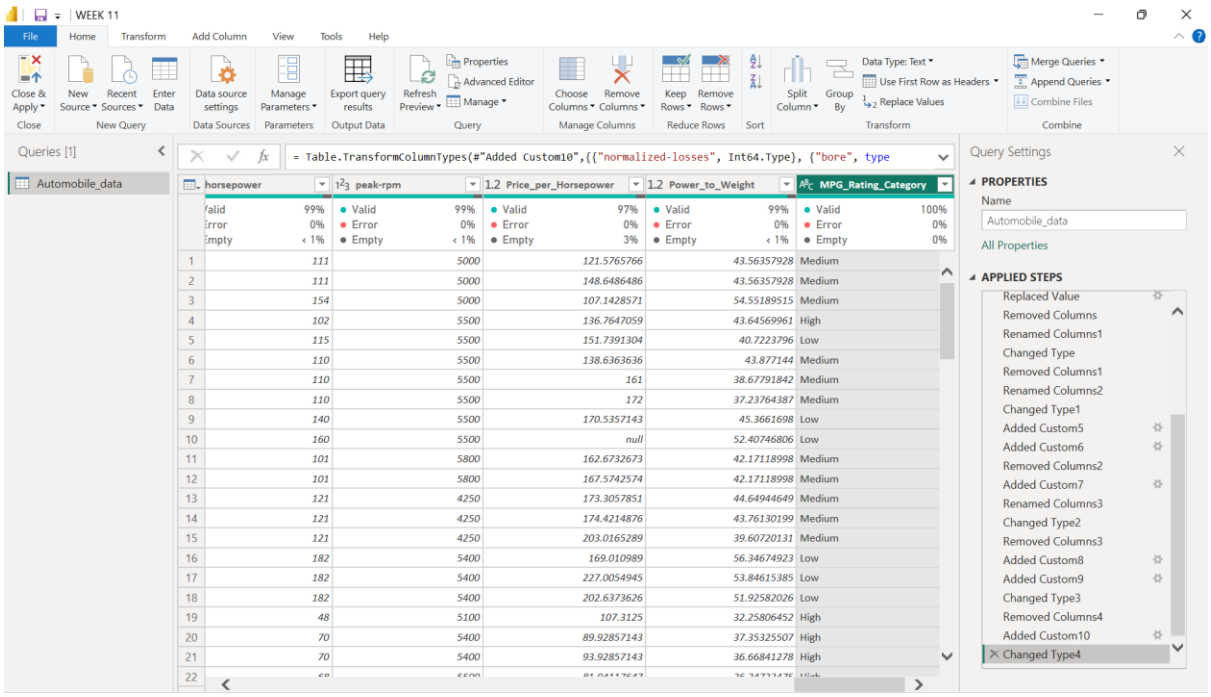
After performing the cleaning, transformation, and feature engineering operations, the dataset was prepared for further analysis.

The final data preparation process included:

- Handling missing and invalid values.
- Correcting numerical data errors.
- Removing unnecessary original columns.
- Checking duplicate records.
- Assigning suitable data types.
- Creating new calculated columns.
- Preparing DAX measures for statistical analysis.

The transformed dataset provides a structured foundation for creating interactive Power BI reports and dashboards.





12. OUTCOME OF THE PROJECT

Through this project, the raw automobile dataset was examined and processed using Microsoft Power BI.

Several data cleaning techniques were applied to improve data quality and consistency.

New calculated columns were developed to provide additional insights into vehicle pricing, engine performance, and fuel efficiency. The project also provided practical experience in using Power Query Editor and DAX formulas.

The final dataset is intended to support further automobile data visualization and analytical reporting.

13. CONCLUSION

The Week 11 Automobile Data Cleaning and Transformation project provided practical knowledge of data preprocessing using Microsoft Power BI.

The project helped in understanding how to identify missing values, handle invalid entries, correct data type errors, check duplicate records, and create meaningful calculated columns through Feature Engineering.

It also improved practical knowledge of Power Query Editor, Custom Columns, data transformation, and DAX measures.

Overall, this project demonstrates the importance of data cleaning and preparation in the data analysis process. A properly transformed dataset helps produce more accurate results and meaningful visualizations in future Power BI dashboard projects.